

Total No. of Questions : 8]

SEAT No. :

PE-2470

[Total No. of Pages : 2

[6584]-449

B.E. (Mechanical / Automobile)/(Honors)

Modelling and Simulation of EHV

(2019 Pattern) (Semester - VII) (402034MJ)

Time : 2½ Hour]

[Max. Marks : 70

Instructions to the candidates :

- 1) Solve Q.No.1 or Q.No.2, Q.No.3 or Q.No.4, Q.No.5 or Q.No.6, Q.No.7 or Q.No.8,
- 2) Figures to the right indicate full marks
- 3) Draw the neat sketch wherever necessary.

- Q1)** a) Explain Torque and speed coupling in electric vehicle? [4]
b) Explain the importance of Sensor Management and Integration for electric vehicle with functional and application point of view? [8]
c) What is Battery/Cell Control System in EV? Explain with its functions and working principal? [8]

OR

- Q2)** a) Explain A Motor Control Unit in electric vehicle with its functional features? [4]
b) Explain EV and EHV configuration based on power electronics? [8]
c) Explain Electronic Control Unit of electric vehicle with its types and working principal? [8]
- Q3)** a) Explain EV configuration and list out the components involved in it? [8]
b) Explain the important feature of following (any four), [8]
i) Unicycle
ii) Bicycle
iii) Dicycle
iv) Tricycle
v) Quadracycle

P.T.O.

OR

- Q4)** a) Explain front and rear wheel drive for electric vehicle with its advantages and disadvantages? [8]
- b) Explain Propulsion and Power distribution system in electric vehicle with its. main components? [8]
- Q5)** a) Explain Chassis frame layout with suitable sketch? Also list out various types of Loads on the Chassis frame? [8]
- b) What do you mean by the term “DFMEA”? Explain the objective feature that can achieve the in the process of system or equipment design? [8]

OR

- Q6)** a) Explain Frame building Problems? Also list out Types of Frame Damage & How to Spot it? [8]
- b) Explain the following parameters related to vehicle dynamics in detail,[8]
- i) Aerodynamic drag
 - ii) Aerodynamic lift
 - iii) Side force
 - iv) Rolling movement
- Q7)** a) What do you mean by Topology and Topography Optimization, explain with suitable example? [10]
- b) Explain Fatigue analysis. List out The factors which affect the fatigue behavior? [8]

OR

- Q8)** a) Explain the phases involved in Crashworthiness Design along with its important features? [10]
- b) Explain Durability along with Factors Affecting Electric Car Dutability? [8]

